

EE016-3-3-PP1 Project Phase 1 – Investigation in Engineering Project

Chapter 3 : Concept Design and Research Methodology

Topic Learning Outcomes

At the end of this topic, you should be able to:

1. Propose the research methodology with concept design derived from fundamental engineering principles by incorporating public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations. (C6, PLO3).
2. Explain professional engineering practices in developing solutions by incorporating social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, with an inclusion of at least eight United Nations Sustainable Development Goals (UN-SDGs). (C5, PLO6)
3. Explain project management execution, economic decision making and entrepreneurial considerations for the proposed solution. (C5, PLO10)

Topic Learning Outcomes

At the end of this topic, you should be able to:

4. Support continuous professional development by recognizing the need for independent and life-long learning with an emphasis on cutting-edge technologies through critical thinking. (C5, PLO11)
5. Justify your work done effectively and professionally through report writing and oral presentation by taking into account cultural, language, and learning differences such as Visual learners, Auditory learners, Kinesthetic learners, Social learners, Solo learners. (A3, PLO9)

Contents & Structure

- Investigation on Tools and Techniques
- Proposed Methodology
- Concept Design derived from Fundamental Engineering Principles
- Professional Engineering Practices
- Project Management, Finance and Entrepreneurship 1

Recap From Last Lesson

- Nil

Key Terms You Must Be Able To Use

If you have mastered this topic, you should be able to use the following terms correctly in your assessments:

- Investigation
- Proposed Methodology
- Concept Design
- Professional Engineering Practices
- Project Management
- Finance
- Entrepreneurship

CHAPTER 3 – CONCEPT DESIGN AND RESEARCH METHODOLOGY

Outline

- i. Introduction
- ii. Investigation on Tools and Techniques
- iii. Proposed Methodology
- iv. Concept Design derived from Fundamental Engineering Principles
- v. Professional Engineering Practices
- vi. Project Management, Finance and Entrepreneurship 1
- vii. Summary

I. INTRODUCTION

- Give a brief background about the chapter.

II. INVESTIGATION ON TOOLS AND TECHNIQUES

- Identify what materials or components are required to implement the project.
- Compare the various types of materials or components available and properly justify which material or component will be used for implementation.
- A table form with explanation is good.
- Avoid using downloading the comparison table from online.
- Don't include any figures/images of materials / components / instruments / devices.
- Don't include any data sheets or architecture or pin diagram from online. You may include any data sheet, if necessary in the Appendix.

III. PROPOSED METHODOLOGY

- Propose the methodology with the help of flowchart or block diagram with explanation.
- The methodology must not be in general rather must be specific to the proposal and must be narrowed.
- By incorporating public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations.

IV. CONCEPT DESIGN DERIVED FROM FUNDAMENTAL ENGINEERING PRINCIPLES

- Give the assumptions considered.
- Tabulate the design specifications.
- Design the necessary parameters, by showing step by step calculation with the equations involved. Do reference the equations, if taken from books or journal / conference papers.
- Tabulate the designed parameters.
- Give the 2D/3D sketch of the design with dimensions.
- Give the GUI developed with explanation.
- A table form with explanation is good.
- Explanation is required wherever possible.

V. PROFESSIONAL ENGINEERING PRACTICES

- Explain the professional engineering practices, standards, rules, laws for safety, health, social, cultural, and legal responsibilities in developing solutions for complex engineering problem.
- By incorporating social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, with an inclusion of at least eight United Nations Sustainable Development Goals (UN-SDGs).
- Avoid explaining in general.

VI. PROJECT MANAGEMENT, FINANCE AND ENTREPRENEURSHIP 1

- Explain how project is managed during Phase I with the help of a Gantt Chart.
- Similarly explain how system implementation, Phase II will be done with timeline, with the help of a Gantt Chart.
- Explain the estimate cost analysis with the help of table.
- Further, explain how well the project in Phase I is managed well.
- Explain how the project can be marketable.
- Avoid explaining in general.

VII. SUMMARY

- Summarize the entire chapter.

QUICK REVIEW QUESTIONS

- Why tools and techniques need to be investigated?
- What is Proposed Methodology?
- What must be covered under concept design?
- What must be related to Professional Engineering Practices?
- What are the five elements of Professional Engineering Practices?

SUMMARY OF MAIN TEACHING POINTS

- Investigation of tools and techniques
- Proposed Methodology
- Concept Design
- Professional Engineering Practices
- Project Management, Finance and Entrepreneurship

QUESTION & ANSWER SESSION

Q & A

WHAT WE WILL COVER NEXT

- Mid-point Oral Presentation